



Automatic seam detection and tracking system for robots based on laser vision[☆]

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ABSTRACT

Detecting weld seams and accurately locating them from time-series images that contain strong noise pollution are difficult and lead to reduced tracking accuracy. To address this problem, this paper presents a seam detection and tracking algorithm from the perspective of seam feature extraction and seam detection and positioning. The proposed image-processing algorithm employs the powerful feature expression capability and self-learning function of the deep convolutional neural network. A real-time weld seam searching and positioning strategy based on the multi-correlation filter cooperative detection mechanism is proposed in consideration of the continuity of the motion of the feature points of adjacent frames and the correlation of laser stripe structural information. Experimental results show that the sensor's measurement frequency can reach 20 Hz, the average absolute tracking error of straight or curved welds is less than 0.25 mm, and the maximum tracking error does not exceed 1 mm in an environment with strong arc and splash noise. Moreover, the welding torch end runs smoothly during welding. The proposed strategy can meet high-quality welding requirements.

1. Introduction

The coordinated development of various technologies, such as robotics and information technology, has led to the unsuitability of manual welding methods with many drawbacks for modern manufacturing applications and their gradual replacement with automated welding systems based on welding robots. However, most robots remain in the fixed mode of “manual teaching–memory reproduction” when performing welding operations due to the current level of robot intelligence. An inevitable deviation exists between the actual trajectory of the weld and the teaching trajectory in an unstructured welding environment due to such factors as thermal deformation, vibration displacement, and clamping error. Welding robots in this working mode cannot perform corresponding corrective actions according to the change in trajectory, thus affecting the final welding quality and accuracy [1]. The most effective means to solve these problems is to develop a smart welding system with environmental information perception and self-cognition functions, and automatic seam-tracking technology is the basic technology and key link for realizing such an intelligent welding system. Automatic seam-tracking technology based on laser vision combines machine vision and structured light 3D vision measurement technology, and it has the advantages of non-contact measurement, accurate feature description, rich perceptual information, and highly autonomous semantic interpretation. Therefore, automatic seam-tracking technology

has elicited increasing attention from local and international scholars [2,3,4]. The working principle of this technology lies in its “sensing–decision–execution” model. Dynamic change information of the weld position is collected in real time by deploying a laser vision sensor in the welding field, and the corresponding control strategy adjusts the welding posture of the robot and the welding torch position flexibly to perform an efficient, automatic, self-adaptive intelligent welding process.

Most researchers have utilized traditional graphic image-processing methods, such as filter denoising, target segmentation, and morphological processing, to extract weld feature points from images acquired during the welding process. These methods extract feature points from the perspective of laser stripe geometry or statistical characteristics of gray value distribution, and a machine-learning method is usually integrated. When the feature information is relatively stable, the method of detecting the target object on the basis of prior knowledge can accurately locate the weld position. However, in the actual environment with unexpected noise interference, only a few algorithms can accomplish the task of automatically and effectively extracting weld feature points. Considering the inter-frame correlation and the smoothness of the laser fringe motion between adjacent frames (i.e., the characteristics of the laser stripe shape and position are not mutated), several scholars have attempted to improve the performance of the weld detection algorithm by using the target visual tracking method in computer

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vision. This method consists of two main stages. The first stage, which involves feature extraction of the input image, aims to extract effective visual features that encode the information of the weld seam and has a low computational load. The second stage, which is the construction of the appearance model, localizes the weld seam in each frame. This method provides a new perspective for designing a robust and effective method for the positioning of the weld seam. Gao et al. [5] proposed a novel method for tight butt joint weld detection, in which the rough location of the weld seam can be determined based on an optical flow measurement method. A particle filter was used to predict the weld position accurately. The advantage of the proposed algorithm is that it can accurately identify and track micro-gap welds with close-to-zero widths or are invisible to the naked eye, but it usually incorrectly regards noise as a detection object. To address the problem of overspending in time and space caused by an excessively large weld seam positioning search area in the traditional algorithm, Li et al. [6] proposed the use of a target-tracking algorithm to determine a small-scale target search area from which the weld seam feature point can be extracted. A specific algorithm was given around the two points, but the algorithm still did not break through the traditional algorithm framework to determine the position of the feature point by using laser stripe geometry. Zou et al. [7] proposed a novel weld seam-tracking algorithm that does not require the extraction of the center of the laser fringe. This algorithm models target state variables in images acquired by a laser vision sensor in a Bayesian probability framework, uses local cosine similarity to find the region with the highest degree of similarity to the image selection module, and introduces a strategy of online updating of discriminant weights to improve the target discernment capability. Experiments showed that the algorithm can accurately extract the feature points of a weld seam image in operation. Later on, Zou et al. [8] designed a robust weld seam tracker by using the morphological image-processing method and a continuous convolution operator tracker. Multi-resolution hand-crafted features called histograms of oriented gradients (HOGs) were extracted from multi-channel feature maps to localize the tracked seam accurately. However, the laser stripe on the image collected by the sensor during the welding process was inevitably blocked or even distorted by splashes and arcs. These previously established methods that utilize traditional hand-crafted features to depict the appearance information of the weld seam perform poorly. Moreover, the specific effects of these methods rely on the knowledge of domain experts to a great extent, and the detection process remains unseparated from people's participation or supervision. Therefore, new methods must be studied from the perspective of improving the robustness and intelligence of their respective algorithms.

Deep learning has developed rapidly in recent years and has resulted in many typical deep convolutional neural network models, such as AlexNet, GoogLeNet, and VGGNet. These models and their derived network models have been widely used in fault diagnosis, speech recognition, human face recognition, and other pattern recognition fields [9–12]. Considering the powerful feature expression capability and self-learning function of deep convolutional neural networks, we first extract features from an acquired image through the feature automatic learning capability possessed by VGGNet's deep structure and make full use of the extracted deep hierarchical features in the appearance of welds. Second, we use a relevant filter with high positioning accuracy and good adaptability to the change in target appearance. A weld joint positioning strategy based on the multi-correlation filter cooperative detection mechanism is proposed. This strategy enables the algorithm to accurately position the weld seam in a feature image coupled with a large amount of noise. Finally, we guide the robot to complete the welding task through further deviation analysis and processing. Experiments show that the system can realize real-time automatic tracking of welds, accurately and reliably detect the position of welds, and achieve smooth welding from different degrees of noise-contaminated sequence images. The final welding accuracy is 0.25 mm. Unlike the majority of current automatic seam-tracking systems that can achieve an accuracy range of

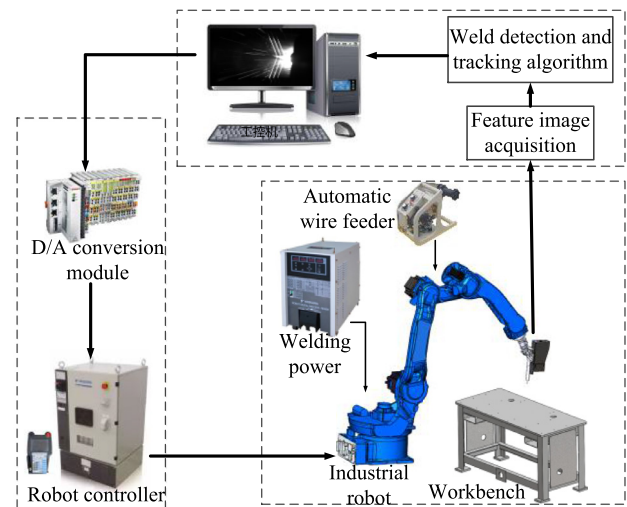


Fig. 1. Schematic of the seam-tracking system.

0.3 mm to 0.5 mm at the most, the proposed system provides more performance advantages and can meet high-quality welding requirements. The paper has the following three main contributions: (1). The proposed image-processing algorithm employs the powerful feature expression capability and self-learning function of the deep convolutional neural network. The limitations of traditional pixel-level feature analyses, single geometric feature recognition, and statistical decision-making methods are overcome by using the deep convolutional neural network structure to extract high-level depth abstraction features of the target. (2). A real-time weld seam searching and positioning strategy based on the multi-correlation filter cooperative detection mechanism is proposed in consideration of the continuity of the motion of the feature points of adjacent frames and the correlation of laser stripe structural information. (3). A series of algorithm verification experiments were conducted on a weld seam tracking experiment platform consisting of a six-axis welding robot, a line laser sensor and an industrial computer. Moreover, the results of the experiment were analyzed from multiple angles.

2. System composition and related working principle

2.1. System and measurement sensor structure

The weld seam automatic tracking system designed in this study consists of image acquisition, image information processing, and seam tracking control modules, as shown in Fig. 1. After the welding starts, the laser vision sensor acquires characteristic images that carry the weld seam position and shape information, which are transmitted to the weld seam feature information processing module through Gigabit Ethernet in real time. The pixel position information is extracted using the weld seam detection and tracking algorithm to reconstruct the 3D spatial information by calibration-related parameters, calculate the deviation between the actual trajectory of the weld and the expected trajectory, and transmit it to the weld seam-tracking controller to adjust the torch travel path at the end of the robot in real time. Online tracking of the weld trajectory is then achieved.

The laser vision sensor is composed of three line laser modules, high-speed industrial cameras, and filters and is in accordance with a specific spatial arrangement, as shown in Fig. 2. A narrow-band interference filter with a center wavelength that is equal to the laser wavelength is installed at the camera lens to appropriately reduce the effects of natural light and arc light on image quality. A heat-resistant, impact-resistant, high-transmittance polycarbonate plate is installed at the lower end of the sensor, and the internal components are protected from damage by insulating the welding heat and splashing without affecting the feature image acquisition. The relative angle between the inclined laser

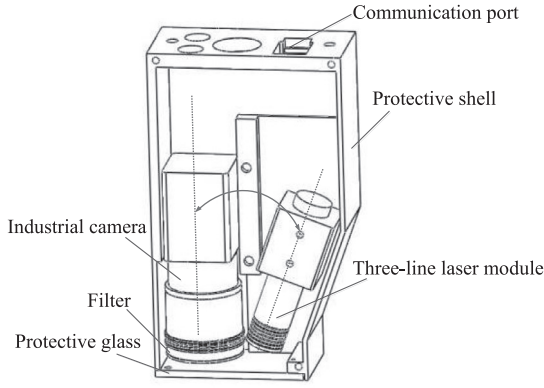


Fig. 2. Diagram of the laser vision sensor.

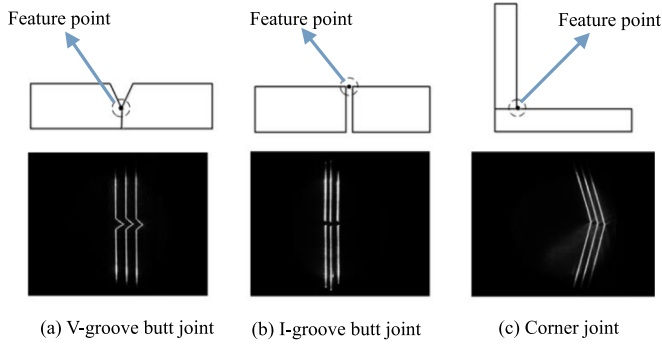


Fig. 3. Characteristic point and corresponding characteristic image of several typical forms of weld.

module and the vertically photographed camera is θ . When the laser beam emitted by the laser module is projected onto the workpiece surface, the beam undergoes a specific deformation at the weld seam and is modulated by the camera to form a characteristic fringe image with hidden weld type and position information. The reason we selected the three-line laser is that the three feature stripes formed by this laser can provide more information for determining feature points than a single feature stripe, especially in the case of the discontinuity of feature stripes caused by noise interference, such as arc light and splash. The three-wire laser is also more robust in feature retention and is thus conducive for accurately extracting the position of the weld feature points. Fig. 3 shows several weld groove types for the weld feature points and the corresponding feature images. For the corner joint, the turning point of the laser stripe corresponds to the bottom center of the weld bevel and is often selected as the characteristic point for weld detection and tracking.

2.2. Working principle of the visual measurement system

The most basic and critical part of the entire system is the visual measurement system composed of laser vision sensors and a robot. Its basic working principle is that after feature points are detected and the pixel coordinates of the captured image are extracted, the perspective projection imaging model is combined with the camera's internal and external parameters and the laser plane calibration results [13]. The mapping relationship between the target 2D image feature information and the 3D position information in the camera coordinate system is then established. The model is then combined with hand-eye calibration of the hand-eye relationship matrix between the camera and the robot, and the spatial coordinates of the feature points in the robot's base coordinate system are determined.

A global coordinate system $O_W X_W Y_W Z_W$, a camera coordinate system $O_C X_C Y_C Z_C$, an imaging coordinate system $O_I X_I Y_I Z_I$, and a pixel co-

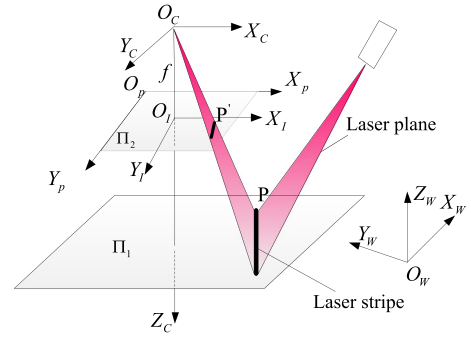


Fig. 4. Perspective projection imaging process of the sensor.

ordinate system $O_P X_P Y_P Z_P$ are established. Fig. 4 shows the perspective projection imaging process of the sensor. The 3D Cartesian space coordinates in the camera coordinate system $O_C X_C Y_C Z_C$ of any point P on the laser stripe formed by the intersection of the laser plane and the object point plane Π_1 are set to (x_c, y_c, z_c) and then act on the optical imaging system. The corresponding image point P' at the coordinate (u, v) on the image-sensitive surface Π_2 can be obtained. According to ideal pinhole imaging theory, the abovementioned projection geometric law can be expressed as:

$$s \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1/f \end{bmatrix} \begin{bmatrix} x_c \\ y_c \\ z_c \end{bmatrix} = \mathbf{M}_1 \begin{bmatrix} x_c \\ y_c \\ z_c \end{bmatrix} \quad (1)$$

where $s > 0$ is a projection scaling factor, f is the focal length, and $\mathbf{M}_1$ is a perspective projection matrix.

The dominant radial distortion in lens distortion is considered, and the distorted image is corrected using the LENZ model as follows [14]:

$$\begin{bmatrix} \tilde{u} \\ \tilde{v} \end{bmatrix} = \frac{2}{1 + \sqrt{1 - 4\kappa(u^2 + v^2)}} \begin{bmatrix} u \\ v \end{bmatrix} \quad (2)$$

where κ is a nonlinear distortion factor that characterizes the degree of radial distortion. The coordinates of the corrected continuous image point are discretized, converted into pixel coordinates (c, r) in the pixel coordinate system $O_I X_I Y_I Z_I$, and stored with the pixel cell array as follows:

$$\begin{bmatrix} c \\ r \\ 1 \end{bmatrix} = \begin{bmatrix} 1/S_x & 0 & C_x \\ 0 & 1/S_y & C_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \tilde{u} \\ \tilde{v} \\ 1 \end{bmatrix} \quad (3)$$

where S_x and S_y are physical dimensions (mm/pixel) of the unit pixel in the horizontal and vertical directions, respectively, which determine the imaging resolution; (C_x, C_y) is the pixel coordinate of the main point formed by the intersection of the optical axis and the image-sensitive surface.

Two constraints on the relative positional relationship between the 2D coordinates of the weld feature points and the spatial 3D coordinates can be established from Eqs. (1)–(3), but the constructed underdetermined equation has insufficient parameters. Consequently, the conversion relationship between the two cannot be uniquely determined, and additional constraints must be provided. The relative positions of the laser module and the camera in the laser vision sensor remain unchanged, and the weld feature points are included on the laser plane. A priori constraint condition is thus established through the condition that the position of the laser plane is constant in the camera coordinate system, i.e.

$$Ax_c + By_c + Cz_c + D = 0 \quad (4)$$

where the vector $\mathbf{n} = (A, B, C)$ is the normal vector of the laser plane and D is the offset distance of the laser plane from the datum plane.

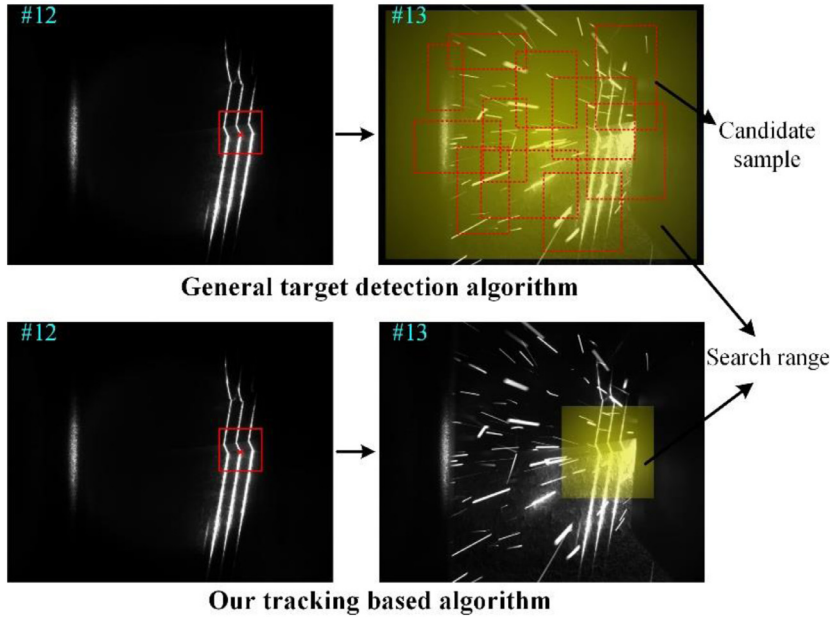


Fig. 5. Search range under different methods.

Eqs. (1)–(3) are combined, and a relationship can be obtained as follows:

$$\begin{cases} x_c = \frac{D(C_x S_x - S_x c)}{A(S_x c - C_x S_x) + B(S_y r - C_y S_y) + C f \epsilon} \\ y_c = \frac{D(C_y S_y - S_y r)}{A(S_x c - C_x S_x) + B(S_y r - C_y S_y) + C f \epsilon} \\ z_c = \frac{D f \epsilon}{A(S_x c - C_x S_x) + B(S_y r - C_y S_y) + C f \epsilon} \end{cases} \quad (5)$$

where $\epsilon = k(C_x^2 S_x^2 + C_y^2 S_y^2 + S_x^2 c^2 + S_y^2 r^2 - 2C_x S_x^2 c - 2C_y S_y^2 r + 1)$, the parameter set $\{f, \kappa, C_x, C_y, S_x, S_y\}$ comprises the camera internal parameters, and $\{A, B, C, D\}$ includes the laser plane parameters.

3. Seam detection and tracking algorithm

3.1. Positioning of the initial weld feature point

The position information of the feature point in the initial frame is crucial to the extraction of the weld features and accurate positioning of the subsequent frames. It is the first prerequisite for seam detection and tracking. The initial image frame in the un-arc-ing state is unaffected by the large random noise. Such noise is only present in the welding state. Coupled with the filtering effect of the narrow band-pass filter, the signal-to-noise ratio and contrast of the image are relatively high. No reference features are used directly for the search and location of feature points; instead, digital image-processing methods are adopted to accurately locate the feature points. The feature image of a lapping fillet weld is used as an example for analysis. We focus on the processing of sequential images. Unlike object detection, we do not need to find the target location on the entire image (1024×1280 pixels). The position of the weld seam frame to frame in video sequence changes very slightly, usually within five pixels. Therefore, the current frame centered at the previous exemplar is cropped as the search image. The size of defined region of interest (ROI) is 300×300 pixels. Since the weld features were relatively simple, the cropped image was able to fully contain the characteristics of the welding seam. As we can see in Fig. 5, the proposed seam-tracking algorithm on sequential images is able to find the target location in a small search range, while others on the entire image. This manner can further simplify the computation. “#12” in the upper left corner of the image denotes 12-th frame.

In this scenario, the tracking algorithm was first initialized at an image region that defines the initial welding feature point. This step is shown in Fig. 6. The tracking algorithm must then crop the original image as the input of the VGGNet when searching for the target object in subsequent frames. The ROI is extracted from the original image captured by the sensor according to the pixel distribution characteristics of structured light stripes in the weld image. First, an appropriate image-filtering method is used to suppress most of the noise in the original observed image, which ensures that the edge details of the laser stripe are maintained as much as possible. Second, threshold processing is applied according to the characteristic that a large difference exists in the gray level between the laser stripe and the background. The purpose is to classify the objects in the image and achieve the separation of laser stripes and backgrounds. Third, binary morphology processing is performed to trim the fringes and eliminate the interference of isolated points and line segments. Fourth, skeleton extraction is performed by selecting the correct connected domain to achieve effective extraction of the laser stripe centerline. Fifth, a line detection algorithm is applied to fit the straight line of the skeleton line, which obtains two fitting lines. Finally, the fitting data and characteristic parameters are combined, and the exact position of the initial weld feature point is obtained through the intersection operation.

3.2. Seam detection and position tracking algorithm

Weld seam detection and tracking realizes real-time extraction and automatic tracking of weld seam position information from image sequences. However, in the welding process, various uncertain factors caused by the complexity of the welding environment, such as surface reflection, arc, and splash, can easily cause the characteristic information in the observed image to be weak or even submerged in strong background noise. Most feature detection and extraction methods used in the seam-tracking system have the disadvantages of insufficient description capability and low discriminability. Consequently, extracting accurate and unambiguous features of this image and detecting the position of the target point are difficult. How to properly describe the target information we are interested in on the basis of the characteristics of the image itself remains a challenging task in the field of current seam tracking; this task is crucial for extracting features with strong distinguishing capability and generalization performance from background noise, solving the problem that the traditional detection algorithm relies on the traversal of the entire image pixel that results in a large space-time overhead,

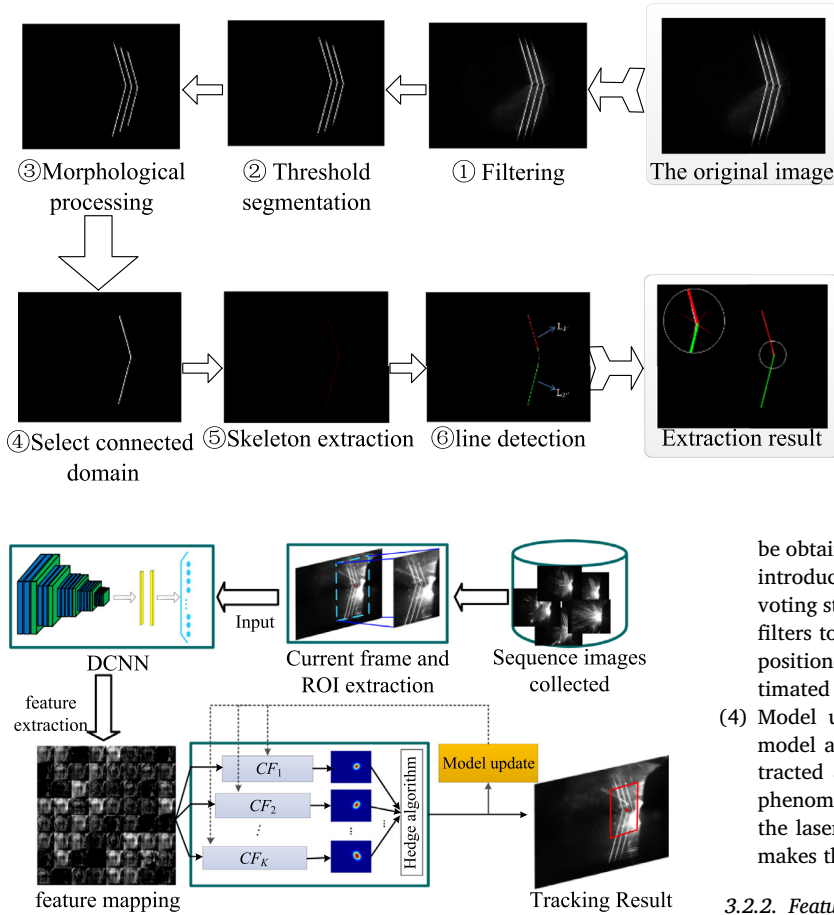


Fig. 6. Extraction process of the initial welding feature point.

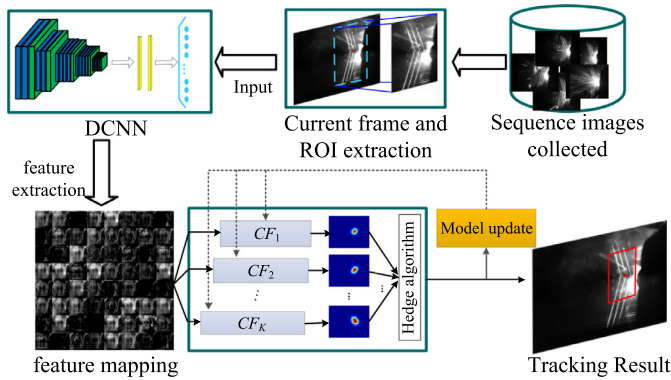


Fig. 7. Algorithm implementation framework.

and achieving robust and rapid detection and precise positioning of the seam from images containing intense noise contamination [15].

3.2.1. Algorithm framework

The inter-frame correlation and the smoothness of laser stripe motion between adjacent frames are considered. A laser stripe in a small area that centers on the feature point serves as a target in the $t-1$ -th frame, and feature expression and description are performed for it. An adaptive feature search is then performed in the t -th frame by constructing a similarity measure or loss function. The position with the largest correlation or the smallest loss value is taken as the new tracking position. The framework of the seam detection and tracking algorithm proposed in this study is shown in Fig. 7, and it includes the following steps.

- (1) ROI extraction. After real-time acquisition of the t -th feature image by the laser vision sensor, a certain rectangular ROI centered at the seam tracking position P_{t-1} in the $t-1$ -th frame is taken as the detection region image to reduce the problem of excessive space and time caused by the significantly large search area in the weld location.
- (2) Feature extraction. A pre-trained convolutional neural network on a large-scale dataset is used as a feature extractor, and feature information is extracted from the slice image to be processed. A feature layer set from a low layer to a high layer can be obtained, which effectively describes the information of the image area distribution.
- (3) Online seam detection and tracking. Correlation operation is first performed between the extracted deep hierarchical features and the correlation filter bank $G = \{CF_1, CF_2, \dots, CF_K\}$ learned at time $t-1$ ($t \geq 1$). Second, each correlation filter obtains the corresponding target confidence distribution map, and the peak position of the confidence map is used as the position of the weld feature point predicted by the filter at time t . The predicted position set $\{O_t^1, O_t^2, \dots, O_t^K\}$ can

be obtained, where K is the number of correlation filters. Finally, by introducing the adaptive hedging algorithm, a multi-expert weighted voting strategy is used to fuse the prediction results of each group of filters to make the final decision. The weld seam can be accurately positioned from the time-series images with noise, and the final estimated weld location O_t can be obtained.

- (4) Model update. The relevant parameters of the correlation filter model are updated online using the deep hierarchical features extracted at the tracking result O_t to effectively suppress the noise phenomenon and the drift caused by the apparent deformation of the laser fringe caused by the deformation of the weld seam. This makes the algorithm robust against drift and noise.

3.2.2. Feature extraction

Li et al. [16] incorporate correlation filters with the fully-convolutional Siamese network to achieve distractor-aware tracking. A deep hierarchical architecture model with the optimal parameter set can be obtained after supervised training of the convolutional neural network through a large number of image samples with different patterns and annotations. This model has excellent feature extraction and description capabilities and can obtain the structural features of the target from the low-level description of the image's detailed information to the high-level reflection of advanced semantics.

Classical convolutional neural network models include LeNet, AlexNet, GoogLeNet, VGGNet and ResNet. Trade-off between the complexity and feature extraction capabilities of the model and motivated by the effectiveness of the VGGNet in the ImageNet Competition, this paper automatically extracts the deep abstract features of the image on the basis of the VGGNet-E network model in the VGGNet series, which has been pre-trained offline on the ILSVRC 2012 Contest universal dataset. ILSVRC 2012 uses a subset of ImageNet of around 1000 images in each of 1000 categories. In all, there were roughly 1.2 million training images, 50,000 validation images and 150,000 testing images. The weights of each layer of the VGGNet were initialized by a Gaussian distribution with an average value of 0 and a variance of 0.01. Some important hyperparameters of the VGGNet include batch size, momentum, weight decay, dropout and initial learning rate. The batch size was set to 256. The momentum was set to 0.9. The weight decay was set to 0.0005. The dropout was set to 0.5. The initial learning rate was set to 0.01. The parameters of VGGNet are listed in Table 1. In the simulation experiment stage, we found that the VGGNet trained by this data set has sufficient recognition strength for seam images, and the effect was much better than the traditional weld tracking algorithm. Seam images has relatively few semantic information and the features are relatively simple. When seam images are used to fine tune the VGGNet, the effect is not obvious, and this may affect the generalization performance of the model. For this reason, the pre-trained model is directly applied to seam images.

Table 1
The parameters of VGGNet.

Layer	Conv1	Conv2	Conv3	Conv4	Conv5	Conv6	Conv7
Parameters	$64 \times 3 \times 3$ st. 1, pad 1	$64 \times 3 \times 3$ st. 1, pad 1 $\times 2$ pooling	$128 \times 3 \times 3$ st. 1, pad 1	$128 \times 3 \times 3$ st.1, pad1 $\times 2$ pooling	$256 \times 3 \times 3$ st. 1, pad 1	$256 \times 3 \times 3$ st. 1, pad 1	$256 \times 3 \times 3$ st. 1, pad 1
Layer	Conv8	Conv9	Conv10	Conv11	Conv12	Conv13	Conv14
Parameters	$256 \times 3 \times 3$ st. 1, pad 1 $\times 2$ pooling	$512 \times 3 \times 3$ st. 1, pad 1	$512 \times 3 \times 3$ st. 1, pad 1	$512 \times 3 \times 3$ st. 1, pad 1	$512 \times 3 \times 3$ st. 1, pad 1 $\times 2$ pooling	$512 \times 3 \times 3$ st. 1, pad 1	$512 \times 3 \times 3$ st. 1, pad 1
Layer	Conv15	Conv16	Conv17	Conv18	Conv19		
Parameters	$512 \times 3 \times 3$ st. 1, pad 1	$512 \times 3 \times 3$ st. 1, pad 1 $\times 2$ pooling	$4096 \times 7 \times 7$ dropout	$4096 \times 1 \times 1$ dropout	$1000 \times 1 \times 1$ softmax		

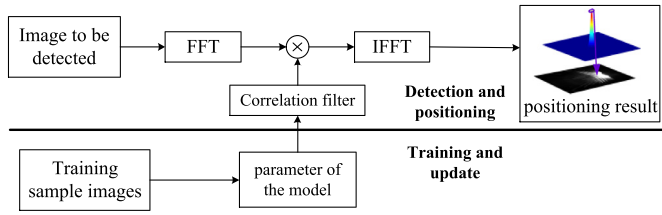


Fig. 8. Principle of correlation filter.

The laser vision sensor acquires the sequence image $\{0, 1, \dots, t\}$ from the start of the welding to the current moment t . A hierarchical feature extraction set $F = \{f_1, f_2, \dots, f_N\}$ that contains dense features can be extracted using the pre-trained VGGNet-E to perform operations, such as convolution and pooling, on the ROI of the current frame t . The set describes the appearance model of the target in the original image from different angles and levels. The convolutional neural network is affected by the downsampling operation in the forward propagation process, and its feature expression capability and discriminative power are continuously improving. The spatial resolution of the feature extraction map is reduced from the lower layer to the higher layer, layer by layer, thereby causing the precise positioning capability of the target to gradually weaken. The size of defined ROI is 300×300 pixels. The Conv3, Conv4, and Conv5 layers of the VGGNet are used to extract features. The width M and height N of input training sample images is 224×224 pixels. The low-level layers extract features at a high spatial resolution. They contribute to localization accuracy. The high-level layers extract features encoding the semantic information. They provide robustness. This raises the question of how to fuse convolutional features to benefit from their complementary properties. Therefore, a multi-correlation filter cooperative detection mechanism is proposed in this paper. It can comprehensively utilize the feature information of the Conv3, Conv4, and Conv5 layers of the VGGNet for detection, and achieve an adaptive balance of spatial information and anti-noise capability. Therefore, balancing the amount of spatial information and anti-noise capability and reasonably selecting the feature extraction layer to detect the weld seam position are necessary.

3.2.3. Detection principle of single correlation filter

The correlation filter is a classifier that can distinguish between the background and the target, and it is generally used in complex scenes in pattern recognition to automatically detect and locate the target [17]. Li et al. [18] present a Coarse-To-Fine Tracking framework which tackles the task of visual tracking with a shallow convolution operator. The principle of the correlation filter is shown in Fig. 8. In the figure, FFT denotes fast Fourier transform, and IFFT denotes inverse fast Fourier transform. A large number of sample images are used in the training phase to learn the spatiotemporal context information of the target, and

a filter model that reflects the mapping relationship between the sample and target position is established in the frequency domain. In the detection stage, a correlation operation is performed between the image to be detected and the learned filter in the frequency domain. A target confidence distribution is formed after inverse Fourier transform. The filter response output map assigns higher confidence to the location where the target appears, and the confidence of the other regions is suppressed. High confidence indicates a high probability that the corresponding location has a target and vice versa.

The method of randomly sampling and acquiring training samples is abandoned when updating the relevant parameters of the filter model to ensure real-time performance during online tracking of welds. Instead, after obtaining the tracking results of the current image frame, a large-scale sample image-training filter is obtained using a cyclic shift strategy [19]. The input training sample image is assumed to be $x_{0,0} \in \mathbb{R}^{(M \times N) \times C}$, where the superscripts M , N , and C denote the width, height, and number of channels, respectively. $x_{0,0}$ is used as the reference sample to perform cyclic shift in vertical and horizontal directions to obtain a new sample image $x_{m,n}$, where $(m,n) \in \{0, 1, \dots, M-1\} \times \{0, 1, \dots, N-1\}$. A dense sample set $\Omega = \{(x_{0,0}, y_{0,0}), \dots, (x_{m,n}, y_{m,n}), \dots, (x_{M-1,N-1}, y_{M-1,N-1})\}$ with labels that obey a 2D normal distribution is then obtained. The probability density function for label $y_{m,n}$ can be described as:

$$y_{m,n} = \frac{1}{2\pi\sigma^2} \exp\left(-\frac{(m-M/2)^2 + (n-N/2)^2}{2\sigma^2}\right) \quad (6)$$

where the average value $(M/2, N/2)$ denotes the position where the reference sample has the highest response value on the confidence map after filtering, and the parameter σ denotes the width of the Gaussian kernel, which determines the sharpness of the ideal confidence map.

For a given sample image, the basic idea of the training correlation filter is to find the optimal filter from all possible solutions so that the difference between the actual and desired outputs is minimal, i.e., the minimum of experience risk, which can be expressed as a solution to the following convex optimization problem with l_2 constraints:

$$\min_w \sum_{m,n} \|w \otimes x_{m,n} - y_{m,n}\|_2^2 + \lambda \|w\|_2^2 \quad (7)$$

where $\otimes$ is the correlation operator, which is defined as the inner product with a linear form in the Hilbert space, i.e., $w \otimes x_{m,n} = \sum_{c=1}^C (w^c)^T x_{m,n}^c$; $\lambda \geq 0$ is a regularization parameter designed to prevent overfitting and is used to weigh the importance of the loss and regularization terms.

The case of a single-channel input (i.e., $C=1$) is considered, and Eq. (7) can be written in a matrix form as follows:

$$\min_w \|\mathbf{X}w - \mathbf{y}\|_2^2 + \lambda \|w\|_2^2 \quad (8)$$

where $\mathbf{X} = [x_{0,0}, \dots, x_{m,n}, \dots, x_{M-1,N-1}]^T$, $x_{m,n}$ denotes a training sample and $\mathbf{y}$ is a label vector that characterizes the desired output. The partial

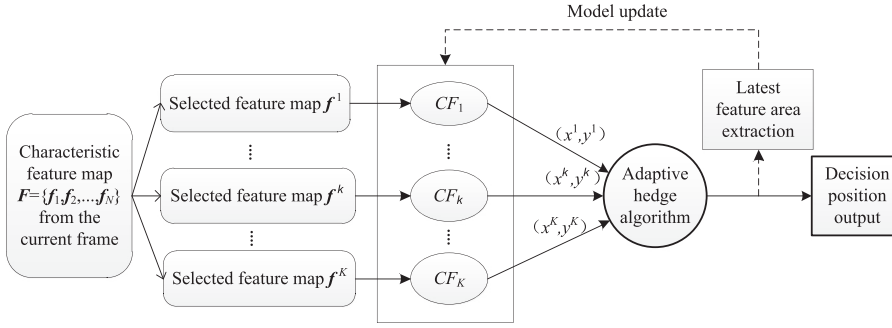


Fig. 9. Seam-positioning framework based on the multi-correlation filter cooperative detection mechanism.

derivative of the parameter vector w in Eq. (8) is calculated, and the derivative is set to zero. The closed-form solution in the complex field can be determined as:

$$w = (X^\dagger X + \lambda I)^{-1} X^\dagger y \quad (9)$$

where $\dagger$ indicates the conjugate transpose and I indicates the unit matrix of the adaptation specification. Given that X is obtained by cyclic shifting of the reference sample $x_{0,0}$, diagonalizing X in the Fourier transform domain yields

$$X = F \text{diag}(\hat{x}_{0,0}) F^\dagger \quad (10)$$

where $\hat{x}_{0,0}$ is the discrete Fourier transform of $x_{0,0}$. Each element in the discrete Fourier transform (DFT) matrix $F \in \mathbb{C}^{s \times s}$ can be defined as:

$$F_{jk} = \frac{1}{\sqrt{s}} \varepsilon^{(j-1)(k-1)} \quad (11)$$

where $\varepsilon = e^{-2\pi i/s}$.

Plugging Eq. (10) into Eq. (9) results in

$$w = F \text{diag}\left(\frac{\hat{x}_{0,0}^*}{\hat{x}_{0,0}^* \odot \hat{x}_{0,0} + \lambda}\right) F^\dagger y \quad (12)$$

Performing DFT simultaneously on both sides can produce the optimal filter in the Fourier transform domain [20]

$$\hat{w} = \frac{\hat{x}_{0,0}^* \odot \hat{y}}{\hat{x}_{0,0}^* \odot \hat{x}_{0,0} + \lambda} \quad (13)$$

where the score means that the numerator and denominator are divided by the elements in the matrix.

In most cases, the input detection image or feature map includes multiple channels (i.e., $C > 1$), such as RGB images, HOG descriptors, and feature maps. In this case, the optimal correlation filter in each channel can be calculated separately as:

$$\hat{w}^c = \frac{\hat{x}_{0,0}^{*c} \odot \hat{y}}{\sum_{c=1}^C \hat{x}_{0,0}^{*c} \odot \hat{x}_{0,0}^c + \lambda} \quad (14)$$

where C is the total number of channels. The algorithm in the Fourier transform domain of Eq. (14) can be used to optimize the computational complexity and large space-time-consuming matrix operations in Eq. (9). In particular, large-scale matrix inversion operations are avoided through the simple Hadamard product operation among matrices that is used in the solution process. This fast solution method provides important technical support for real-time target detection.

After the optimal correlation filter w is obtained through training, target detection can be performed on a new image frame. The input detection feature image is assumed to be $z \in \mathbb{R}^{(M \times N) \times C}$, and the target detection formula is

$$r_p(w, Z) = Zw \quad (15)$$

where Z is a cyclic matrix based on z . Direct calculation is complicated. Therefore, we make optimizations. First, after FFT is applied to each channel of z , a corresponding $\hat{z}^c$ can be obtained, and a multiplication

operator that corresponds to the elements between $\hat{z}^c$ and corresponding filter $\hat{w}^c$ is performed. Second, after summing the results of each channel by summing and using IFFT, the final output filter response map can be obtained as:

$$r = F^{-1} \left(\sum_{c=1}^C \hat{w}^c \odot \hat{z}^c \right) \quad (16)$$

3.2.4. Seam positioning based on the multi-correlation filter cooperative detection mechanism

In this study, instead of directly inputting the image to be detected or performing a simple artificial feature extraction, a feature map automatically extracted by the deep convolutional neural network is inputted. The use of a single correlation filter to filter a single feature map does not guarantee the robustness and accuracy of target detection in complex environments. Therefore, a multi-correlation filter cooperative detection mechanism is proposed based on a single correlation filter. An adaptive hedge algorithm [21] that combines the detection results of all correlation filters to predict the final position is also introduced. The framework is shown in Fig. 9. This idea is similar to the application of ensemble learning that constructs a strong classifier from a number of weak classifiers. Synthesizing the weld detection results of each correlation filter to predict the final position is a complex system decision problem, and its essence is to use appropriate information-processing methods to integrate or select the information given by multiple decision experts to obtain the best decision results. The most commonly used method is the voting method, which assigns different weight coefficients to each expert according to the value of the given information and eventually fuses all the information to make decisions. The key to this method is how to determine a reasonable weight coefficient and update the method.

For the k -th correlation filter CF_k ($k = 1, 2, \dots, K$) in the correlation filter bank, the detected image it receives is a feature map extracted from the l -th convolutional layer of the deep convolutional neural network. Both deconvolution and bilinear interpolation can be used for up-sample the feature. Deconvolution is not adopted for the following two reasons. First, some research results show that the deconvolution does have good feature reduction ability. However, the seam tracker used in this paper is not an end-to-end mode. That is, the feature extraction module (VGGNet part) and the similarity matching module (correlation filter part) exist independently. If the deconvolution is used, it means that the deconvolution layer needs to be added before the similarity matching module. However, the deconvolution requires offline training, which is different from the online training mechanism of the correlation filter. This will cause the deconvolution to be untrained. Second, the Conv3, Conv4, and Conv5 layers of the VGGNet are used to extract features in this paper. The three-layer features can generate images of the same size by deconvolution with a certain step size. However, if the step size of the deconvolution is set too large, the features of each layer can generate the features of the original image size after one deconvolution, but the final detection accuracy is extremely low. Although the detection accuracy will be improved, the detection speed will be greatly reduced

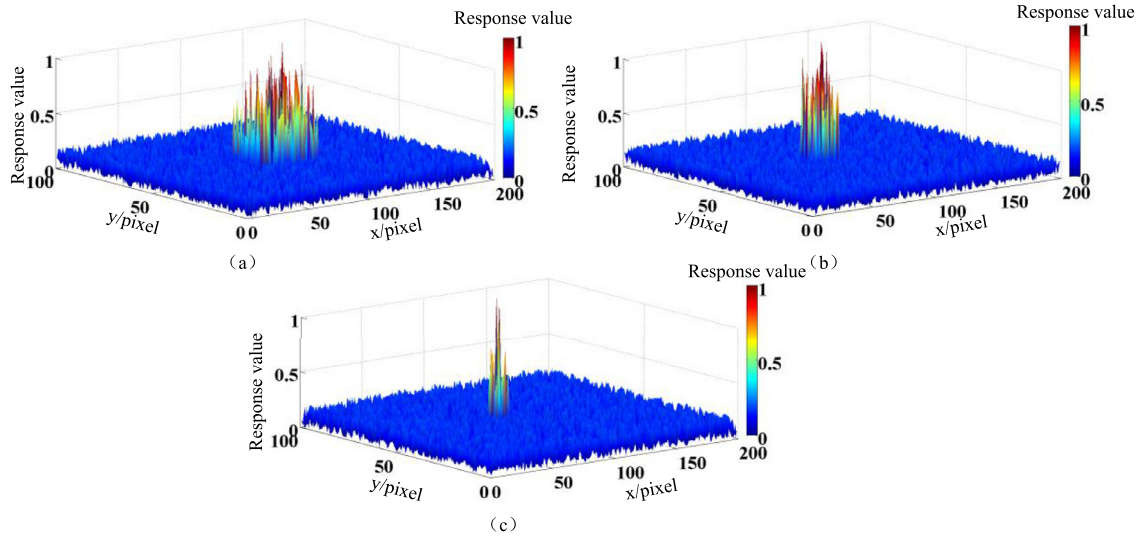


Fig. 10. Response output maps of different correlation filters.

if multiple deconvolution operations with the step size of 2 are carried out. Considering the real-time and accuracy requirements of weld inspection, bilinear interpolation is used to up-sample the feature map and alleviate the issue that the dimension of the feature map is reduced layer by layer due to the pooling operation. Thus, the input of each filter is a fixed size

$$\mathbf{g}_i = \sum_j \kappa_{ij} \mathbf{f}_j \quad (17)$$

where $\mathbf{g}$ denote the up-sampled feature map and $\mathbf{f}$ denote the feature maps before up-sampling. The interpolation weight κ_{ij} depends on the distance between pixel positions i and j . The response output map of CF_k can be calculated through Eq. (16). Fig. 10 shows several response output maps of the correlation filter from high level to low level. Fig. 10(a)–(c) correspond to Conv5, Conv4, Conv3 layers of pre-trained VGGNet respectively. According to the criterion that the maximum response point corresponds to the position where the target has the highest probability of occurrence, the positioning result O^k of CF_k is

$$(x^k, y^k) = \arg \max_{x', y'} r^k(x', y') \quad (18)$$

where (x^k, y^k) are the coordinates of the target position detected by the correlation filter CF_k .

Let $\{O_t^1, O_t^2, \dots, O_t^K\}$ denote the position set obtained from parallel detection in the t -th frame of the correlation filter bank. The correlation filter bank $G = \{CF_1, CF_2, \dots, CF_K\}$ is considered a decision group, where any one of the correlation filters CF_k is an independent decision-making expert. The overall decision-making positioning results O_t can be obtained through the voting mechanism as:

$$(x_t^*, y_t^*) = \sum_{k=1}^K w_t^k \cdot (x_t^k, y_t^k) \quad (19)$$

where K denotes the total number of decision experts and w_t^k corresponds to the weight coefficient of each expert, which is subject to $\sum_{k=1}^K w_t^k = 1$. At the initial moment (i.e., $t = 1$), each expert assigns the same weight $w_t^k = 1/K$ a priori, and other moments can be determined by the hedge algorithm after the weld seam is positioned at the previous moment. The specific process is discussed below.

The simplest hedge algorithm is the standard parameter-free hedge algorithm. The main idea is to calculate the confidence difference between each expert's decision position and the overall decision position to obtain the loss value of each expert and to iterate the calculation according to certain criteria. The target is estimated to be hidden in the state space and then assigned to each expert with different weight coefficients

[22]. At time $t > 1$, once the overall decision position $O_{t-1} = (x_{t-1}^*, y_{t-1}^*)$ of the target in the image at the previous time is obtained, the loss value of each expert is

$$l_{t-1}^k = \max(r_{t-1}^k) - r_{t-1}^k(x_{t-1}^*, y_{t-1}^*) \quad (20)$$

where r_{t-1}^k is the filter response output map matrix of the correlation filter at time $t - 1$, $\max$ is used to calculate the maximum of all elements in the operated matrix object, and $r_{t-1}^k(x_{t-1}^*, y_{t-1}^*)$ is the value of the element at coordinate (x_{t-1}^*, y_{t-1}^*) in matrix r_{t-1}^k . At this moment, the total loss value is a weighted combination of all expert loss values:

$$\bar{l}_{t-1}^k = \sum_{k=1}^K w_{t-1}^k l_{t-1}^k \quad (21)$$

where w_{t-1}^k corresponds to the weight coefficient of each expert in the decision phase of $t - 1$ moment.

In the standard parameter-free hedge algorithm, to measure the effectiveness of the information provided by various experts on the decision, the **regret** of each expert is defined as:

$$r_{t-1}^k = \bar{l}_{t-1}^k - l_{t-1}^k \quad (22)$$

The recurrence formula for the cumulative regret value of each expert from the initial time to $t - 1$ can be given as:

$$R_{t-1}^k = (1 - \rho) \cdot R_{t-2}^k + r_{t-1}^k \quad (23)$$

where ρ is the forgetting factor. The cumulative regret value is used as a random variable, and the half-normal distribution is adopted as a prior distribution. We can then obtain the probability density function of regret as:

$$\Phi(x, \theta) = \exp\left(-\frac{([x]_+)^2}{2\theta}\right) \quad (24)$$

where $x \in \mathbb{R}$, $[x]_+$ denotes $\max\{0, x\}$ and $\theta > 0$ is a scale parameter that determines the overall height and steepness of the probability density function. The cumulative regret value R_{t-1}^k subject to a half-normal distribution and the corresponding scale parameter θ_{t-1} satisfy the following constraints

$$\frac{1}{K} \sum_{k=1}^K \exp\left(-\frac{([R_{t-1}^k]_+)^2}{2\theta_{t-1}}\right) = e \quad (25)$$

In the standard parameter-free hedge algorithm, the property that the weight coefficient reassigned to each expert in each phase is proportional to the first derivative value of the cumulative regret value is

Table 2

Weight coefficients computation for use in the expert weighted voting strategy.

Algorithm 1: Weight coefficient w_t^k computation	Description
1: $w_1^k = \frac{1}{K}, R_1^k = 0, \Delta t$	initializes the same weight and cumulative regret value at time period $t = 1$
2: $l^k = [l_1^k] = [0]$	initializes the loss value at time period $t = 1$
4: for $t = 2, \dots, N$ do	
4: $(x_{t-1}^*, y_{t-1}^*) = \sum_{k=1}^K w_{t-1}^k \cdot (x_{t-1}^k, y_{t-1}^k)$	the positioning result at time $t - 1$
5: $l_{t-1}^k = \max(r_{t-1}^k) - r_{t-1}^k(x_{t-1}^*, y_{t-1}^*)$	calculate the loss value at time $t - 1$
6: update $l^k = [l_1^k, l_2^k, \dots, l_{t-1}^k]$	
7: $l_{t-1}^k = \sum_{k=1}^K w_{t-1}^k l_{t-1}^k$	calculate the total loss at time $t - 1$
8: $r_{t-1}^k = l_{t-1}^k - l_{t-1}^k$	calculate the regret at time $t - 1$
9: if $t < \Delta t$	
10: $R_{t-1}^k = (1 - \rho) \cdot R_{t-2}^k + r_{t-1}^k$	
11: else	
12: $\mu_t^k = \frac{1}{\Delta t} \sum_{\tau=t-\Delta t+1}^t l_{\tau}^k, \sigma_t^k = \sqrt{\frac{1}{\Delta t-1} \sum_{\tau=t-\Delta t+1}^t (l_{\tau}^k - \mu_t^k)^2}$	calculate the mean and standard deviation of filter CF_k 's detection loss during time period $[t - \Delta t, t]$
13: $s_t^k = \frac{ l_t^k - \mu_t^k }{\sigma_t^k}$	calculate the cumulative regret of each correlation filter CF_k at time t
14: $\alpha_t^k = \min(g, \exp(-\gamma s_t^k))$, $R_t^k = (1 - \alpha_t^k) \cdot R_{t-1}^k + \alpha_t^k r_t^k$	calculate the cumulative regret of each correlation filter CF_k at time t
15: end if	
16: $\frac{1}{K} \sum_{k=1}^K \exp(-\frac{(R_{t-1}^k)^2}{2\theta_{t-1}}) = e$	calculate the corresponding scale parameter θ_{t-1}
17: $w_t^k = \frac{ R_{t-1}^k }{\theta_{t-1}} \exp(-\frac{(R_{t-1}^k)^2}{2\theta_{t-1}})$	calculate the weight coefficient at time t
18: End For	

proven. After R_{t-1}^k and θ_{t-1} are obtained by Eqs. (23) and (25), the following relationship can be used to determine the weight coefficient that should be assigned to each expert at the time using the above property

$$w_t^k \propto \frac{\partial \Phi(x, \theta)}{\partial x} \Big|_{x=R_{t-1}^k, \theta=\theta_{t-1}} = \frac{[R_{t-1}^k]_+}{\theta_{t-1}} \exp\left(-\frac{([R_{t-1}^k]_+)^2}{2\theta_{t-1}}\right) \quad (26)$$

When the cumulative regret value R_{t-1}^k of a decision maker is less than 0, the corresponding w_t^k equals 0, which is equivalent to neglecting the decision results provided by the expert.

Eq. (23) indicates that in the standard parameter-free hedge algorithm, all the related filters use the same forgetting factor ρ when calculating the accumulated regrets at any time. They directly add all the regrets of the current time and do not consider the influence of the detection stability difference in each moment on the calculation method of the accumulated regrets. This method is effective in the case of small noise interference and smooth movement of the weld seam in images, but it is difficult to apply to automatic seam tracking with complicated working conditions. Therefore, on the basis of the standard parameter-free hedge algorithm, an adaptive hedge algorithm based on the Gauss model is proposed in this study to model and describe the loss value.

The change in the observed value of the target in a short time is small. Hence, the correlation filter CF_k 's detection loss during time period $[t - \Delta t, t]$ is modelled via the Gauss distribution with mean μ_t^k and standard variance σ_t^k form as follows:

$$\mu_t^k = \frac{1}{\Delta t} \sum_{\tau=t-\Delta t+1}^t l_{\tau}^k \quad (27)$$

$$\sigma_t^k = \sqrt{\frac{1}{\Delta t-1} \sum_{\tau=t-\Delta t+1}^t (l_{\tau}^k - \mu_t^k)^2} \quad (28)$$

The stability of each correlation filter CF_k at time t is measured by the following formula:

$$s_t^k = \frac{|l_t^k - \mu_t^k|}{\sigma_t^k} \quad (29)$$

A small s_t^k indicates that the stability of the detection is high. When calculating the cumulative regret value R_t^k , the smaller s_t^k is, the larger the proportion of the current regret value r_t^k is. By contrast, we hope to rely more on the historical regret value and weaken the effect of the

current regret value. On the basis of this principle, the cumulative regret value calculation process is adaptively adjusted via

$$R_t^k = (1 - \alpha_t^k) \cdot R_{t-1}^k + \alpha_t^k r_t^k \quad (30)$$

$$\alpha_t^k = \min(g, \exp(-\gamma s_t^k)) \quad (31)$$

where $0 \leq \alpha_t^k < 1$ is the learning factor; γ is the scale parameter; and $g < 1$ is the upper limit of the set ratio, which is used to prevent the excessive value of the learning factor that leads to a low proportion of historical regret values. Using these two equations instead of Eq. (23) helps reasonably assign the weight coefficients of the relevant filters during the seam-tracking process, thereby greatly improving the accuracy of obtaining the best position through the overall decision. The procedure for determining the weight coefficients is summarized in Table 2.

3.2.5. Model update

After the new weld position O_t is obtained, the correlation filter can be updated in real time by using Eq. (14). However, this method only considers the tracking results of the image frames at the current moment, and unsuitable samples are inevitably added to the update process of the filter, resulting in a low degree of fit between the filter and the target and even in failure in the next frame detection. Therefore, the numerator A_t^c and denominator B_t^c of Eq. (14) are updated using the incremental method as follows

$$A_t^c = (1 - \eta)A_{t-1}^c + \eta \hat{x}_{0,0}^{c*} \odot \hat{y} \quad (32)$$

$$B_t^c = (1 - \eta)B_{t-1}^c + \eta \sum_{c=1}^C \hat{x}_{0,0}^{c*} \odot \hat{x}_{0,0}^c \quad (33)$$

$$w_t^c = \frac{A_t^c}{B_t^c + \lambda} \quad (34)$$

where t and c denote the frame index and the feature channel index, respectively; η is a smoothing factor that ensures the smoothness of the update of the correlation filter and avoids abrupt mutations.

3.3. Comparative tests

The comparative tests are conducted by using two feature extraction methods, including Histogram of Oriented Gradients (HOG) feature extraction methods and DCNN (Keep the same ROI algorithm and correlation filter algorithm). The results are given in Figs. 11 and 12. Clearly, the DCNN features provide improved results compared to HOG features.

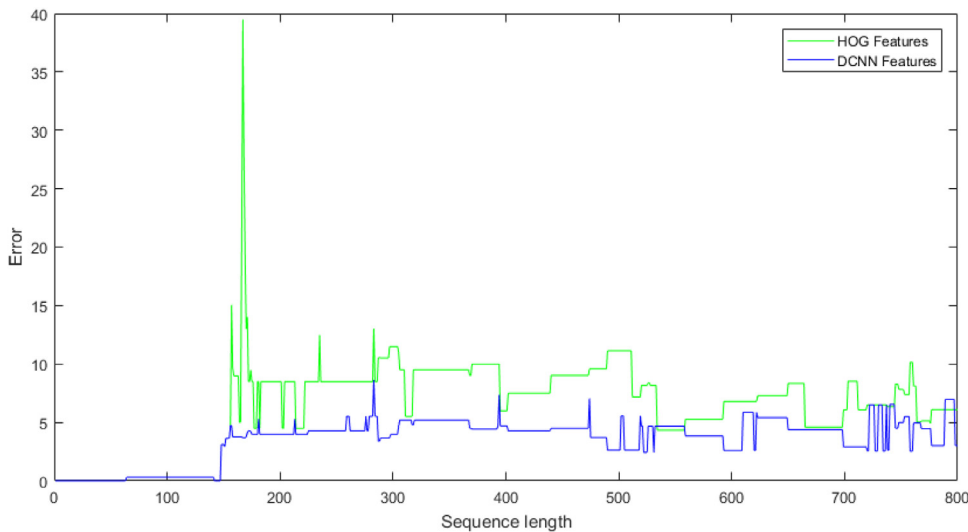


Fig. 11. Tracking error.

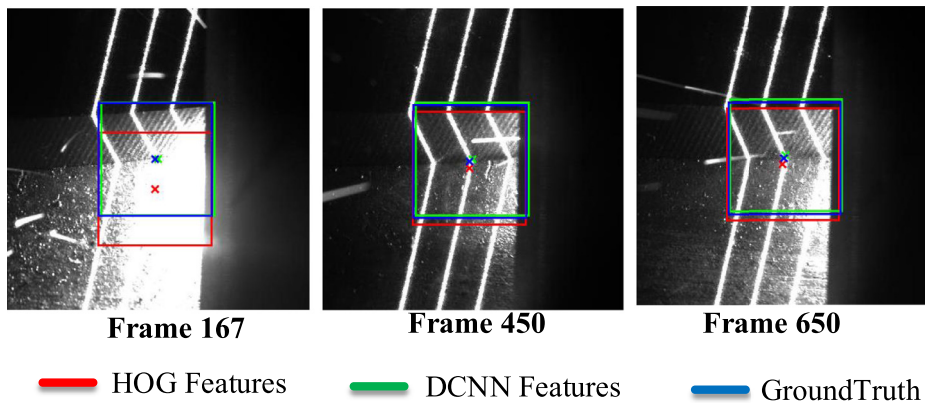


Fig. 12. Test results of the HOG Features (red) and DCNN Features (green).

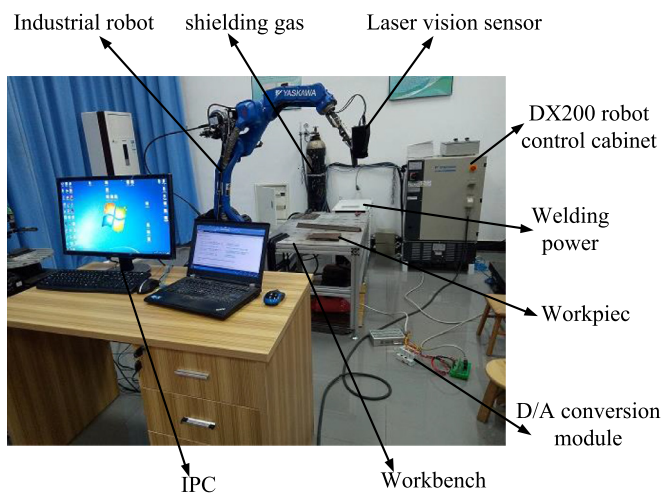


Fig. 13. Physical map of the experimental platform.

4. Experimental research

4.1. Experimental platform

The physical map of the laser vision weld seam automatic tracking experimental platform constructed in this study is shown in Fig. 13. The platform is composed of a laser vision information acquisition module, a

weld seam characteristic information-processing module, a weld seam-tracking control module, and other functional units. The composed hardware devices mainly include laser vision sensors, industrial robots that support welding equipment, high-performance industrial control machines, D/A conversion modules, and various types of wire materials.

The welding robot body is the Yaskawa MOTOMAN-MA1440-C six-axis welding robot, the control cabinet is DX200, the welding power is MOTOWELD-RD350, and the general sensor built in the robot controller is the JANC-D-YCPO2-E type synchronous tracking substrate.

The camera used in the laser vision sensor is Basler's aca1280-60gm-type CMOS camera equipped with an e2v EV76C560 CMOS sensor, with a frame rate of 60 fps and a resolution of 1.3 million pixels. The camera lens is Computar's M0814-MP2 type lens, with a C-mount lens interface, a fixed focal length of 8 mm, an aperture range of F1.4 to F16, and a target surface size of 2/3". The laser module adopts Nanoline Laser's NL-660-100-01L-30 three-wire laser transmitter with a wavelength of 660 nm, a power of 100 mW, and a fan angle of 30°. The protective glass is a polycarbonate plate of the type PCTSP4H-70-40-3-F60 customized by MISUMI with a total light transmittance of 91%. The filter is a narrow-band interference filter manufactured by Edmund Optics with a central wavelength of 660 nm, a bandwidth of 10 nm, and a transmittance of 85% to 95%.

The IPC adopts Advantech's IPC-510 fan-less IPC, an on-board Intel i7-3770 quad-core processor with 3.4 GHz frequency, and NVIDIA GeForce GTX980 GPUs for multi-thread parallel processing of data. The D/A conversion module is used to convert the output digital signal into a range voltage signal that can be accepted by the universal sensor and drive the robot motion to achieve track correction; it mainly includes an

Table 3
The running time of seam detection and tracking algorithm.

Welding object	Number of images/frame	Processing time/s	Mean fps	Running time/ms
Straight V-groove butt weld	356	15.06	23.63	42.32
curved V-groove butt weld	381	16.65	22.88	43.70
S-shaped curve lap fillet weld	372	15.93	23.35	42.82

Table 4
Welding process parameters.

Parameters	Value
Welding method	MIG
Welding voltage(V)	19.8
Welding current(A)	130
Welding speed(mm/s)	5.5
Wire feed speed(mm/s)	5
Wire diameter(mm)	1.2
Wire extension(mm)	10
Welding material	45
Shielding gas	92%Ar+8%CO ₂
Gas flow rate(L/m)	16

EL4134 analog output terminal module, an EK1100 coupler from Beckhoff, a DC 24 V/10A switching power supply, and a distribution line.

The PC software is built on the Visual Studio 2013 integrated development environment using C#, which accesses the functions of HALCON, MATLAB, MotoPlus, and other software libraries to complete the collection of feature images, the detection and tracking of weld feature points in images, the reconstruction of weld feature points in three dimensions, and tracking control. The laser vision sensor is used to obtain real-time welding images with a 20 Hz sampling frequency and 1280pixel × 1024pixel image size.

The proposed image-processing algorithm employs the powerful feature expression capability and self-learning function of the deep convolutional neural network. The convolutional network involves a large number of matrix operations, which strongly affects its real-time performance. Here, we use M_w and M_h to denote the size of the feature maps and K_w and K_h to denote the size of the convolutional filters. l denotes the current convolutional layers. C_{in}^l and C_{out}^l denote the input and output of current convolutional layers. D_c denote the numbers of convolutional layers in the CNN. Thus, the online computational complexity is $O(\sum_{l=1}^{D_c} M_w^l M_h^l K_w^l K_h^l C_{in}^l C_{out}^l)$.

In order to improve system real-time performance, we transferred matrix operations to the GPU (Graphics Processing Unit) via CUDA (Compute Unified Device Architecture). The matrix operations between images and filters could be assigned to GPU under the control of the CPU. Although these calculations are repeated operations, we have to send them into multiple threads due to the characteristic of data structure. Hence we simplified the sliding operation by reconstructing the image and filter matrix. This method could be extended to four-dimensional array, that is, multi-batches feature maps and filters with multi-channels.

As shown in Table 3, the running time of seam detection and tracking algorithm is less than 45 ms, and the control period of robot was set to 50ms. The relevant process parameters in the welding experiment are shown in Table 4.

4.2. Experimental results and analysis

Automatic tracking welding experiments are conducted for welds with different welding groove types and trajectory shapes to verify the actual application effect of the algorithm. Straight welds include straight lap fillet welds and straight V-groove butt welds, and curved welds include curved lap fillet welds and curved V-groove butt welds.

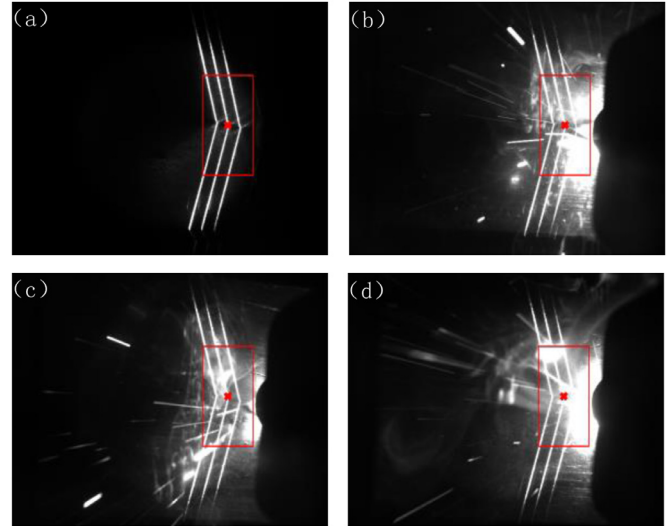


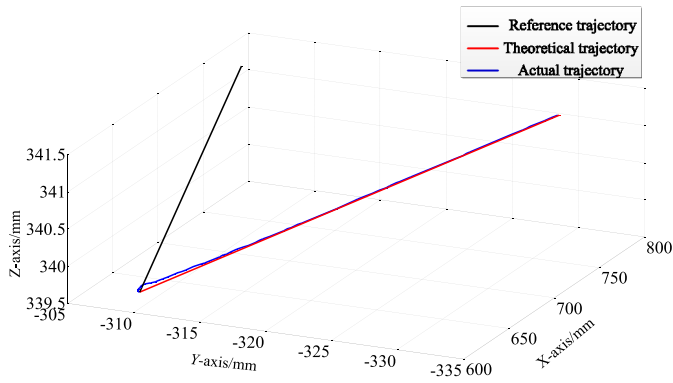
Fig. 14. Images collected and corresponding tracking results (“x”). (a) Initial position and (b)–(d) automatic tracking welding process.

4.2.1. Automatic welding experiment on straight welds

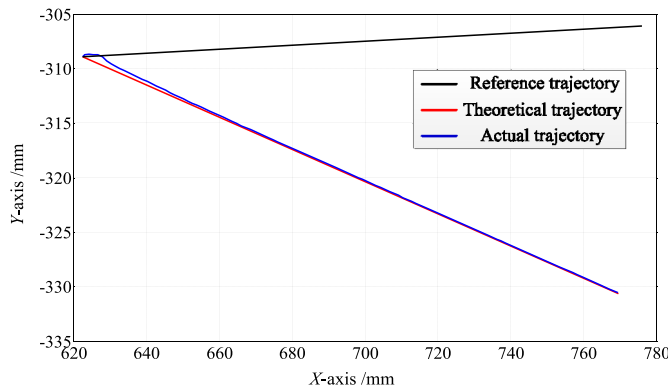
The automatic tracking welding of straight lap fillet welds is performed. Before the experiment, the reference trajectory of the weld to be tracked and the theoretical trajectory after the offset are determined. After the torch is manually adjusted to reach the initial weld position, the first image is obtained with a laser vision sensor, as shown in Fig. 14(a), where the turning point of the laser stripe represents the actual position of the weld. After the image preprocessing method is used to determine the initial position of the weld feature point, a rectangular area centered on the feature point is selected as the tracking target, in which the weld feature point is identified via the symbol “x.” Afterward, an automatic welding program that incorporates all proposed algorithms is initialized, and the robot is continuously controlled to find the actual trajectory of the weld. Fig. 14(b)–(d) show the captured images during the tracking process and the detection and positioning result of the weld seam feature points in each image frame. The actual welding trajectory can be obtained by recording the global coordinates of each sampling point in the process.

The 3D coordinates of the reference trajectory, theoretical trajectory, and actual welding trajectory in the experiment are shown in Fig. 15(a). The projections on the XOY-plane and XOZ-plane are given in Fig. 15(b), (c), where X, Y, and Z directions denote the $O_B X_B$, $O_B Y_B$, and $O_B Z_B$ directions in the robot base coordinate system, respectively. The workpiece is fixedly installed on the worktable parallel to the XOY-plane under the robot base system. Although the table plane and the XOY-plane are not strictly parallel due to machining error, i.e., the coordinates of the weld in the Z direction are not always consistent, the theoretical trajectory almost coincides with the actual trajectory in the Z direction. Therefore, the effective error of seam tracking is only constrained on the XOY-plane. In the subsequent experiments, only the effective error on the XOY-plane is considered by default.

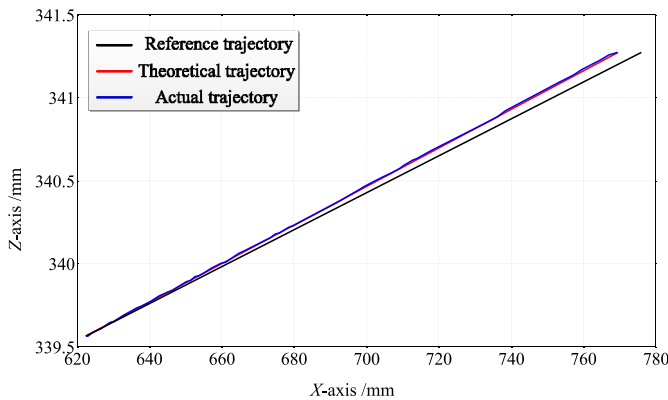
The error curve shown in Fig. 16 can be plotted according to the obtained actual tracking trajectory and the theoretical tracking trajectory. The average absolute tracking error can be calculated by using the



(a) 3D graph of seam-tracking results



(b) Projection diagram on the XOY-plane



(c) Projection diagram on the XOZ-plane

Fig. 15. Tracking results of a straight lap fillet weld.

following formula to measure the tracking accuracy of the weld seam

$$\bar{E} = \frac{\sum_{i=1}^n |E_i|}{n} \quad (35)$$

where n is the total number of samples and E_i is the tracking error at each sampling point. After calculation, the average absolute tracking error is $\bar{E} = 0.1776$ mm, the maximum tracking error during the period is $E_{\max} = 0.6134$ mm, and the root-mean-square error that reflects the tracking process fluctuation is $E_{\text{rms}} = 0.1795$ mm. The welding object is shown in Fig. 17. The results show that although a large amount of noise interferes in the automatic welding process, the weld seam-tracking system proposed in this study can still obtain high welding accuracy. The

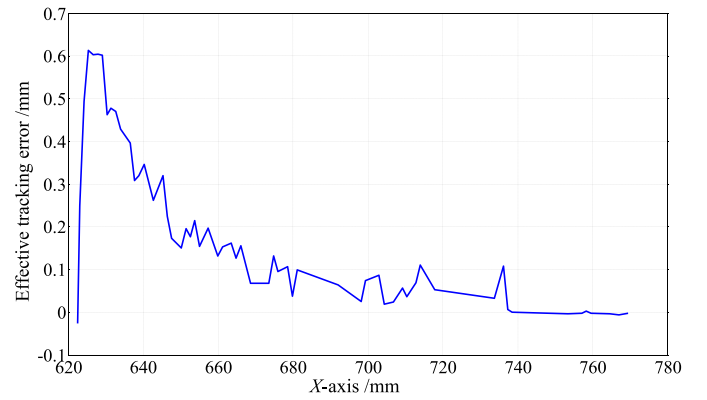


Fig. 16. Effective tracking error of a straight lap fillet weld.



Fig. 17. Welding object of a straight lap fillet weld.

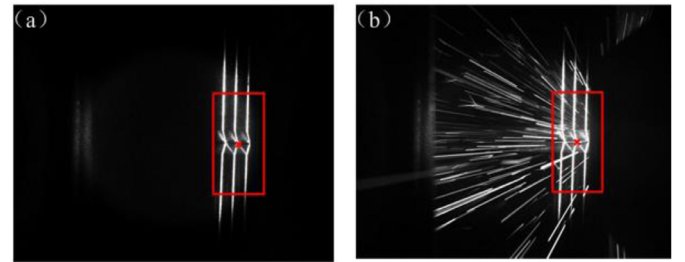
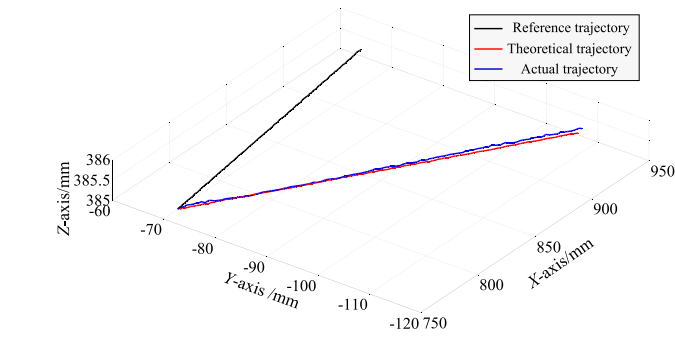


Fig. 18. Images collected and corresponding tracking results ("x"). (a) Initial position and (b) automatic tracking welding process.

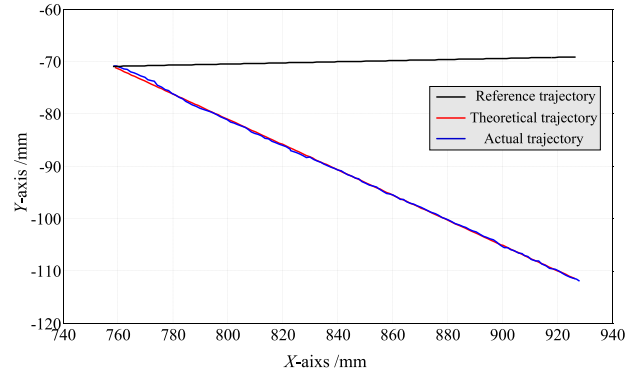
weld bead of the welding material is also even and tightly combined, which fully meets the requirement.

Under the premise of unchanged welding process parameters, an experiment is conducted with a straight V-groove butt weld. The initial feature image and the welding feature image acquired by the laser vision sensor are shown in Fig. 18, and the position of the feature point and the tracking area are automatically identified in each frame. Even when the signal-to-noise ratio of the camera's field of vision is relatively small, it can still perform effective and accurate tracking, and no drift phenomenon occurs.

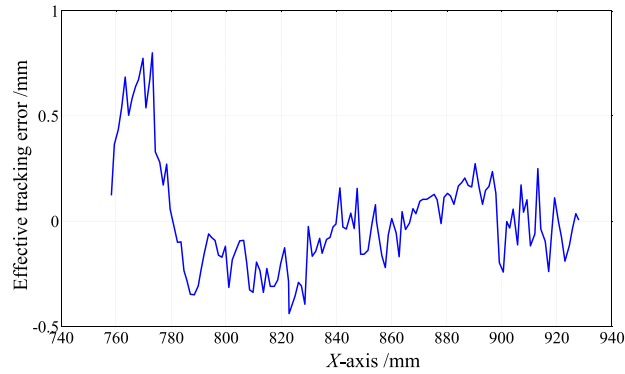
The reference and theoretical welding trajectories of the weld are measured before welding, and the 3D map shown in Fig. 19(a) can be obtained by plotting them with the actual welding trajectory acquired through the automatic tracking of the weld. The projections on the XOY-plane are shown in Fig. 19(b). The theoretical trajectory of the weld before welding and the actual tracking trajectory obtained by automatic welding are compared. Only the effective error constrained to the XOY-plane is considered, and a tracking error graph is created, as shown in Fig. 19(c). After calculation, the average absolute tracking error is $\bar{E} = 0.1872$ mm, the maximum tracking error during the period is $E_{\max} = 0.8078$ mm, and the root-mean-square error that reflects the tracking process fluctuation is $E_{\text{rms}} = 0.2902$ mm. The welding object is shown in Fig. 20.



(a) 3D graph of seam-tracking results



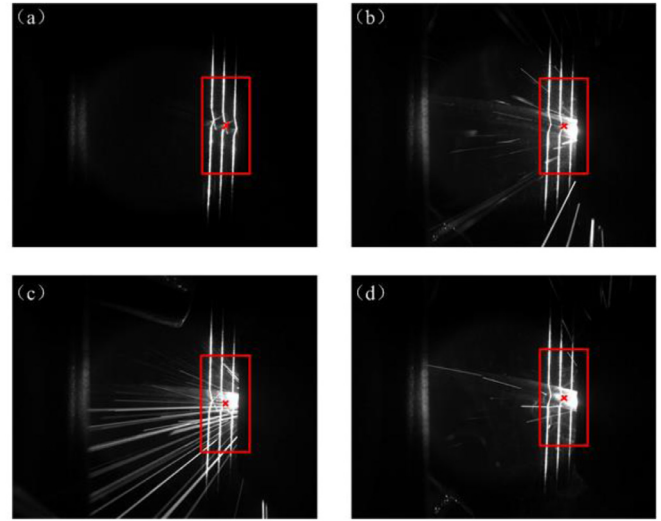
(b) Projection diagram on the XOY-plane



(c) Effective tracking error of a straight V-groove butt weld

Fig. 19. Tracking results of a straight V-groove butt weld.

Comparison of the experimental results of the straight V-groove butt weld with those of the straight lap fillet weld shows that both tracking errors are large in the initial stage of welding. The actual tracking trajectory is biased toward the reference trajectory first and then gradually approaches the theoretical trajectory and fluctuates near it. The root cause of this problem lies in the time-delay effect of the arcing of the welding robot. During this period, the automatic seam-tracking algorithm does not work, the robot moves “inertially” according to the reference trajectory until the tracking algorithm works, and the trajectory is automatically corrected. The average absolute error of the automatic tracking experiment for the straight lap fillet weld is slightly smaller than the experimental results for the tracking of the straight V-groove butt weld. The root-mean-square error of the former’s characterization of volatility is smaller than that of the latter. The error curve depicts that the degree of jitter in the former welding is much smaller than that in the latter. The reason is that the characteristics of the images in the seam tracking of the straight lap fillet weld are more obvious, and the

**Fig. 20.** Welding object of a straight V-groove butt weld.**Fig. 21.** Images collected and corresponding tracking results (“x”). (a) Initial position and (b)–(d) automatic tracking welding process.

algorithm can still accurately detect the position of the weld in the case of noise covered by welding and splashing.

4.2.2. Automatic welding experiment on curved welds

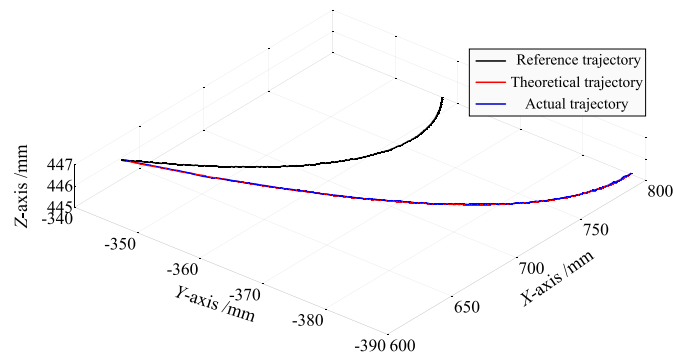
The most obvious difference between curved and straight welds is the large difference in the shape and position of laser stripes in adjacent image frames. Therefore, the weld seam detection and tracking algorithm must have a robust performance and high requirements on the real-time performance of the system. A simple arc-shaped weld is adopted as the curved V-groove butt weld, and the circular and theoretical trajectories after the offset need to be taught using the circular interpolation instruction MOVc before the welding. The images captured by the laser vision sensor during the automatic welding process are shown in Fig. 21.

Similar to the automatic welding experiment on the straight weld, the 3D map, as shown in Fig. 22(a), can be obtained by plotting the reference and theoretical welding trajectories with the actual welding trajectory obtained through the automatic seam tracking. The projections on the XOY-plane are shown in Fig. 22(b). The error calculation involves the geometric equation of the theoretical trajectory; therefore, the theoretical trajectory must be fitted according to the sampling points obtained by the robot. Considering that the curved weld is arc-shaped, according to the least-square method of fitting a circular arc [19], the final expression of the theoretical trajectory can be obtained by performing arc fitting on the sampled data as follows

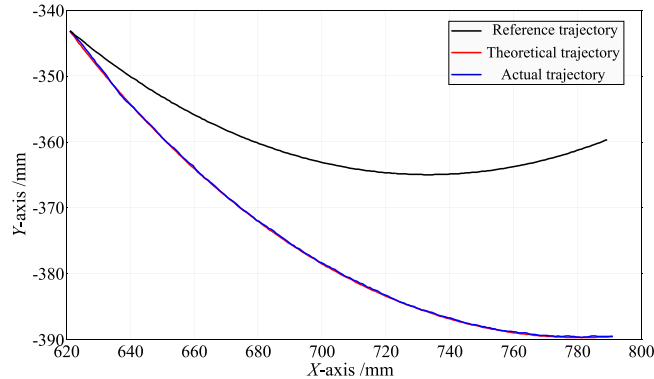
$$(x - 781.1062)^2 + (y + 91.3725)^2 = 298.3585^2 \quad (36)$$

After the actual tracking trajectory is obtained through the experiment, an effective tracking error curve can be drawn according to the geometric equation of the theoretical trajectory, as shown in Fig. 22(c).

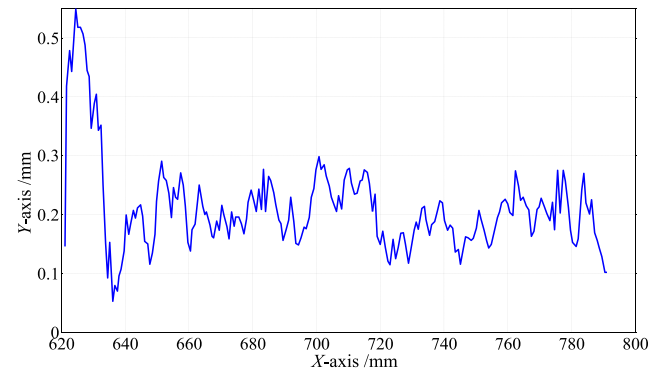
After calculation, the average absolute tracking error is $\bar{E}=0.2129$ mm, the maximum tracking error during the period is



(a) 3D graph of seam-tracking results



(b) Projection diagram on the XOY-plane



(c) Effective tracking error of a curved V-groove butt weld

Fig. 22. Tracking results of a straight V-groove butt weld.

$E_{\max}=0.5490$ mm, and the root-mean-square error that reflects the tracking process fluctuation is $s_E=0.2850$ mm. The welding object is shown in Fig. 23. The error graph demonstrates that similar to the result of the automatic tracking experiment for a straight weld, the initial stage of welding error is also relatively large due to the arc delay effect and then becomes stable in the small value range. In addition, the oscillating amplitudes of the adjacent tracking sample points are relatively small, indicating that the seam-tracking process is relatively stable and that the system meets the smoothness criteria of the curve welding requirements.

An S-shaped curve lap fillet weld formed by the outer cut of two arc segments is specifically made to verify the robustness of the algorithm and validate the effect of the algorithm on arc transition (i.e., whether the algorithm can make adaptive changes according to the changes in the curve and then track the curve robustly). Unlike the curved V-groove butt welds, the radius of curvature of the two arc weld seams that form the S-shaped lap weld seam is much smaller, which means that the tra-



Fig. 23. Welding object of a curved V-groove butt weld.

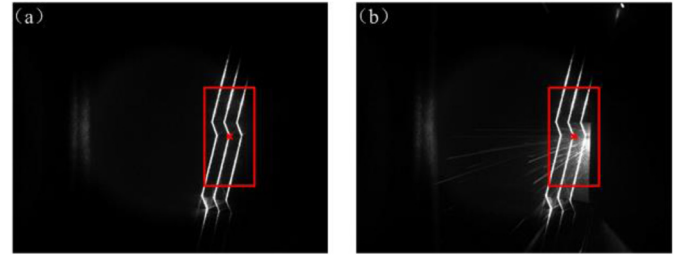


Fig. 24. Images collected and corresponding tracking results ("x"). (a) Initial position and (b) automatic tracking welding process.

jectory tracking is more difficult and the system requirements are more demanding. An experiment is performed after the reference and theoretical trajectories of the weld are confirmed. During the welding process, the sensor continuously collects feature images and extracts feature points to obtain the real-time position of the trajectory, which in turn drives the robot to track the trajectory effectively. Fig. 24 shows the characteristic images in the initial stage of welding and the feature images that contain noise during the welding process.

The 3D coordinates of the reference trajectory, theoretical trajectory, and actual tracking trajectory in the experiment are shown in Fig. 25(a) and projected onto the XOY-plane, as shown in Fig. 25(b), to calculate the effective tracking error.

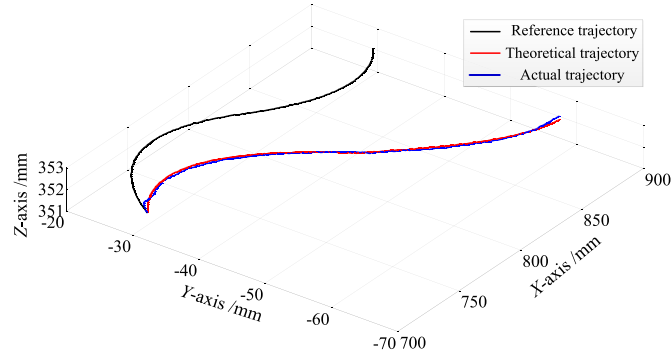
The theoretical trajectory must be fitted to the sample point obtained by robot teaching to calculate the tracking error of the actual trajectory relative to the theoretical trajectory. The S-shaped curve weld consists of two tangent arc welds; hence, it is limited by the "tangential" constraint, which makes it impossible to directly use the statistical regression model for curve fitting. Instead, after determining the position of the tangent point, the least-square fitting model is used to perform arc fitting on the discrete points collected. The specific methods are as follows:

- (1) The total number of discrete data points p_i collected through robot teaching is assumed to be N , and the n -th point p_n is used as the demarcation point. The point set $\Omega = \{p_i | 1 \leq i \leq N \text{ and } i \neq n\}$ can be divided into two groups. The first group contains $n - 1$ points, and the second group contains the remaining $N - n$ points. The point set meets $3 \leq n \leq N - 3$.
- (2) The two sets of measurement points are used to fit the geometric equations of the arcs, and the equations of the two arcs are calculated as F_1 and F_2 according to the least-square arc-fitting model. The mathematical expressions are as follows

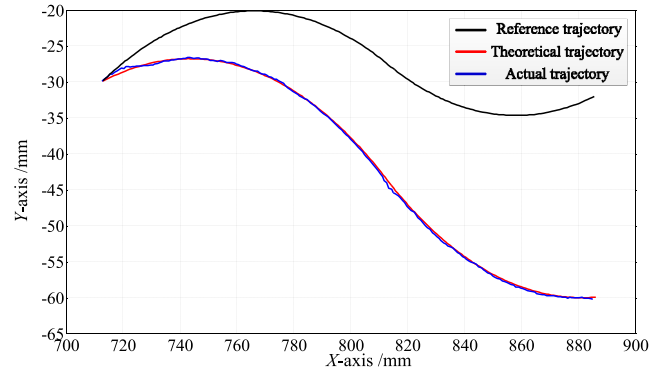
$$(x - x_{o1})^2 + (y - y_{o1})^2 = R_1^2 \quad (37)$$

$$(x - x_{o2})^2 + (y - y_{o2})^2 = R_2^2 \quad (38)$$

where (x_{o1}, y_{o1}) and (x_{o2}, y_{o2}) are the center coordinates of two arcs and R_1 and R_2 are the radii of two arcs.

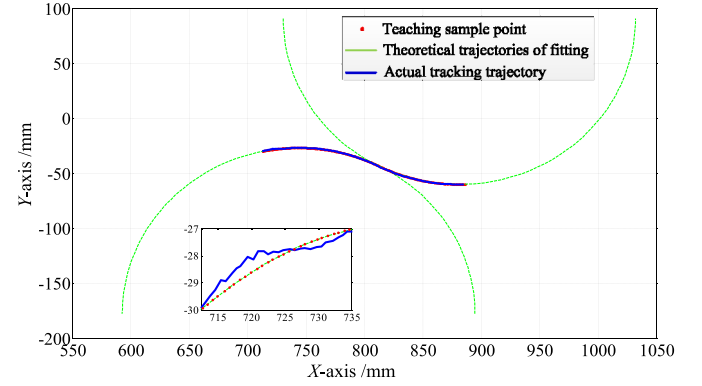


(a) 3D graph of seam-tracking results

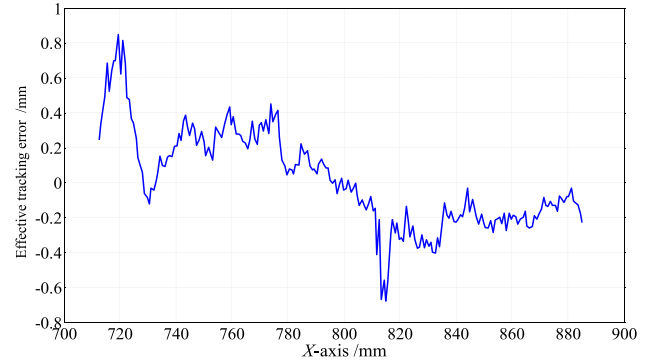


(b) Projection diagram on the XOY-plane

Fig. 25. Tracking results of an S-shaped curve lap fillet.



(a) Comparison of theoretical and actual tracking trajectories



(b) Effective tracking error of an S-shaped curve lap fillet weld

Fig. 26. Weld seam-fitting result and error curve.



Fig. 27. Welding object of an S-shaped curve lap fillet weld.

- (3) The quasi-errors of the arcs of each segment at this time are calculated. The sum of squared errors of all discrete data points $p_i(x_i, y_i)$ ($1 \leq i < n$) on the first segment arc is

$$\Delta_1 = \sum_{i=1}^{n-1} (x_i - x_{o1})^2 + (y_i - y_{o1})^2 - R_1^2 \quad (39)$$

Similarly, the sum of squared errors of the points $p_j(x_j, y_j)$ ($n+1 \leq j \leq N$) on the second arc is

$$\Delta_2 = \sum_{j=n+1}^N (x_j - x_{o1})^2 + (y_j - y_{o1})^2 - R_2^2 \quad (40)$$

The task of finding the point p_n closest to the tangent point can be converted into the problem of minimizing the sum of squared sums of the global fitting error as follows

$$\hat{n} = \arg \min_{3 \leq n \leq N-3} \Delta_1 + \Delta_2 \quad (41)$$

The $\hat{n}$ -th collection point $p_{\hat{n}}$ closest to the tangent point can then be solved.

- (4) Two independent sets of data points can be divided through the above steps. The constraints of two arc excises are added, and the fitting problem can be expressed as a nonlinear optimization problem as follows

$$\begin{aligned} \min \Delta_1 + \Delta_2 \\ \text{s.t. } (x_{o1} - x_{o2})^2 + (y_{o1} - y_{o2})^2 = (R_1 + R_2)^2 \end{aligned} \quad (42)$$

With the fitted arcs F_1 and F_2 as the initial conditions, the final parameters of the two arcs can be solved iteratively.

The total number of data points in this experiment is $N = 218$, and the calculated $n = 127$. The geometric equations of the two arcs obtained by the final fitting are

$$(x - 743.4881)^2 + (y + 177.7240)^2 = 150.9483^2 \quad (43)$$

$$(x - 881.1301)^2 + (y - 90.7109)^2 = 150.7326^2 \quad (44)$$

The comparison between the theoretical and actual tracking trajectories fitted is shown in Fig. 26(a). An effective tracking error graph can be created, as shown in Fig. 26(b). After calculation, the average absolute tracking error is $\bar{E} = 0.2317$ mm, the maximum tracking error during the period is $E_{\max} = 0.8504$ mm, and the root-mean-square error that reflects the tracking process fluctuation is $s_E = 0.2553$ mm. The welding object is shown in Fig. 27. The maximum absolute error of welding shows an abnormal increase near the outer cutting position of two arc weld seams. The reason is that the trajectory changes in the direction of the slope of the outer tangent at this point, and the robot's movement pattern changes drastically, resulting in a large error. However, the change is quickly adjusted to reduce the tracking error. According to the overall experimental results, the proposed algorithm can achieve high tracking accuracy in the automatic tracking welding of fillet welding joints and can meet welding quality requirements in actual welding applications.

4.3. Result analysis

The experimental results of the automatic tracking welding experiments for each group of welds indicate that the average absolute error is less than 0.25 mm, the maximum tracking error does not exceed 1 mm, and the tracking effect of the linear weld is much better than that of the curve weld. Most automatic weld seam tracking systems that are already in the commercial or research stage are limited to the tracking application of a single type of weld. The best actual tracking error is in the range of 0.3 mm to 0.5 mm [13,23,24], and the tracking accuracy of curve weld seam tracking is less satisfactory. Unlike them, the weld detection and localization algorithm proposes a strategy with better robustness and adaptive performance, which makes the system achieve high tracking accuracy and self-adaptive capability in the automatic tracking of straight and curved welds. Without changing the core part of the system, it can be applied to multi-type weld seams.

5. Conclusions

With the aim to solve rapidly detecting weld seams and accurately locating them from time-series images that contain strong noise pollution, a seam detection and tracking algorithm was proposed from the perspective of seam feature extraction and seam detection and positioning in this paper.

The powerful feature expression capability and self-learning function of the deep convolutional neural network are combined. The high-level deep abstract features of the target are extracted through the deep convolutional neural network structure, which overcomes the limitations of traditional pixel-level feature analysis and single geometric feature recognition or statistical decision-making methods and reduces the ambiguity of features; second, a real-time weld seam searching and positioning strategy based on the multi-correlation filter cooperative detection mechanism is proposed by considering the continuity of the motion of the feature points of the adjacent frames and the correlation of laser stripe structure information. This strategy can solve the technical problem of intelligent detection and positioning of welds under large amounts of noise.

A weld seam tracking experiment platform is established, and a weld seam automatic tracking experiment is performed using the proposed algorithm in the actual welding environment. The experimental results show that under strong arc and splash interference, the sensor measurement frequency reaches 20 Hz, the average absolute tracking error is less than 0.25 mm, and the maximum tracking error does not exceed 1 mm, and the welding torch end runs smoothly during welding. In summary, the system can realize real-time tracking of welds, can accurately and reliably detect the position of welds from different degrees of noise pollution images, and can realize smooth welding. The proposed system has more advantages and can meet high-quality welding requirements.

Declaration of Competing Interest

We declare that we do not have any commercial or associative interest that represents a conflict of interest in connection with the work submitted.

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